

BUKGANGNEUNG, Korea, Republic of

WMO: 471040

Lat: 37.8044N

Lon: 128.8553E

Elev: 80

StdP: 100.36

Time Zone: 9.00 (E09)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-7.1	-25.7	0.4	-5.4	-23.6	0.5	-4.7	7.4	-1.4	6.7	-0.2	3.0	160	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	32.2	23.4	30.9	22.9	29.3	22.1	25.2	29.1	24.5	28.4	23.9	27.7	2.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.2	26.9	23.5	18.4	26.6	22.7	17.5	26.1	77.8	29.0	74.9	28.4	72.3	27.7	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.5	5.5	4.7	-12.4	34.3	2.5	1.2	-14.2	35.2	-15.6	35.9	-17.0	36.6	-18.8	37.5		
			-14.0	26.0	2.3	0.9	-15.6	26.6	-17.0	27.1	-18.2	27.6	-19.9	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	0.1	1.7	6.3	12.0	17.6	20.3	24.2	24.4	19.9	15.1	8.7	2.3
	DBStd	9.16	3.68	3.59	4.38	4.28	3.88	2.92	3.35	2.82	2.21	2.92	4.01	4.10
	HDD10.0	1018	308	233	134	27	1	0	0	0	0	3	71	241
	HDD18.3	2578	566	466	374	196	63	13	1	0	9	105	288	497
	CDD10.0	2027	0	1	19	86	234	309	439	447	297	161	33	1
	CDD18.3	547	0	0	1	4	39	72	182	189	56	4	0	0
	CDH23.3	3798	0	0	1	26	268	401	1517	1435	144	7	0	0
	CDH26.7	1127	0	0	0	3	48	83	543	436	14	0	0	0
Wind	WSAvg	2.0	2.3	2.0	2.2	2.4	2.1	1.5	1.6	1.5	1.5	1.8	2.2	2.3
Precipitation	PrecAvg	1403	52	54	70	79	79	111	231	277	220	107	80	39
	PrecMax	2096	248	196	174	248	277	343	758	1137	559	348	348	234
	PrecMin	924	0	0	8	1	3	14	58	55	0	2	0	0
	PrecStd	279	50	48	43	56	55	70	134	178	149	90	75	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	13.7	21.8	25.7	30.3	30.7	33.5	33.8	28.4	24.3	19.8	13.9
		MCWB	4.4	7.0	12.0	13.3	16.6	19.0	23.4	24.8	21.0	15.1	12.6	8.1
	2%	DB	8.6	10.6	18.6	23.6	27.9	28.9	32.4	32.3	26.5	22.3	17.9	11.6
		MCWB	3.3	4.9	10.4	12.4	15.3	18.4	23.5	23.9	20.5	15.4	11.4	6.9
	5%	DB	6.7	8.6	16.2	21.7	26.2	27.3	31.2	30.8	24.8	20.9	16.2	9.9
		MCWB	1.9	3.6	8.4	11.6	15.2	18.2	23.4	23.9	19.7	14.9	11.0	5.3
	10%	DB	5.4	7.1	14.0	19.3	24.6	25.8	29.9	29.1	23.7	19.7	14.7	8.2
		MCWB	1.1	2.8	6.8	10.4	14.3	18.2	23.0	23.2	19.4	14.5	10.1	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	8.3	13.5	15.1	19.7	21.9	25.8	26.1	23.2	19.8	15.0	10.0
		MCDB	9.0	12.7	20.7	23.0	24.1	26.1	30.2	30.3	26.1	20.8	17.9	12.5
	2%	WB	4.2	5.7	10.9	13.9	17.5	21.0	25.1	25.4	22.0	17.9	13.4	7.4
		MCDB	7.0	9.0	17.8	21.7	24.6	25.1	29.5	29.1	24.2	20.1	16.3	11.0
	5%	WB	3.0	4.5	9.1	12.6	16.5	20.3	24.4	24.8	21.1	16.8	12.0	5.7
		MCDB	5.7	7.7	14.8	19.4	23.0	24.2	29.0	28.3	23.3	19.1	14.6	8.9
	10%	WB	1.7	3.4	7.4	11.4	15.7	19.7	23.7	24.2	20.3	15.7	10.6	4.4
		MCDB	4.8	6.4	13.0	16.9	21.6	23.4	28.3	27.6	22.8	18.4	13.9	7.8
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.0	7.5	8.2	8.3	6.4	5.5	5.6	6.2	7.2	7.0	7.3
		MCDBR	8.0	9.0	10.4	11.2	10.8	9.5	7.6	7.7	7.6	8.5	7.9	8.0
		MCWBR	5.5	6.0	5.9	5.6	4.9	3.9	3.1	3.3	4.4	5.1	5.3	6.1
	5% WB	MCDBR	6.7	8.4	9.6	10.2	9.1	7.0	6.5	6.4	5.8	6.3	6.5	7.1
		MCWBR	5.2	6.1	5.9	5.7	5.0	3.4	3.1	3.3	3.8	4.8	5.1	5.9
Clear-Sky Solar Irradiance	taub	0.419	0.484	0.581	0.624	0.629	0.679	0.640	0.568	0.495	0.488	0.452	0.410	
	taud	1.964	1.838	1.680	1.647	1.680	1.642	1.783	1.960	2.085	1.973	1.994	2.018	
	Ebn at Noon	712	713	682	687	695	660	683	723	755	703	673	690	
	Edh at Noon	135	175	226	247	243	252	217	178	149	151	131	120	
All-Sky Solar Radiation	RadAvg	2.41	3.19	4.11	4.91	5.44	5.16	4.29	4.27	3.90	3.41	2.41	2.15	
	RadStd	0.21	0.31	0.41	0.39	0.53	0.43	0.60	0.54	0.47	0.29	0.31	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	-0.88	N/A	N/A	N/A	N/A	N/A	+96	+51

Nomenclature: See separate page